

Lecture 15

Simulation

Business Analytics, 1e

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Learning Objectives

- Concept of simulation
- Applications in Business (Risk management)
- Develop and apply Monte Carlo simulations.

Review of Prescriptive Analytics

- Prescriptive analytics *provides advice on*
“*what businesses should do.*”
 - Quantify the effect of different possible actions to make a more informed decision.
 - We can use **simulation** algorithms to:
 - To imitate a real-world process to produce several scenarios identified by decision-makers.

Assessing Risks

- Risk can be defined as the possibility or probability that an undesirable outcome may occur.
- Risk analysis (also called scenario analysis) refers to a process of examining and assessing different scenarios in order to better understand the possibility of both intended and undesired outcomes.
- We can *use random variables to capture the inherently uncertain nature of processes* to make better-informed decisions.

Assessing Risks

- Examine multiple scenarios with different possible outcomes to assess risk.
- To captures some of the uncertainty of a real-life process, consider at least three scenarios:
 - Most likely
 - Pessimistic (worst case)
 - Optimistic (best case)
- It may be difficult to determine what the best/worst cases are.
- Use historical data to help estimate the best/worst cases.

Assessing Risk

- Example: A company in the western United States manufactures and distributes women's hand-bags to retail stores around the country.
- To manufacture a handbag, the company usually incurs a variable cost of \$40 per handbag, but the actual costs in the past have ranged from \$35 to \$46 per handbag.
- The overall fixed cost is usually \$43,000 but may vary between \$31,000 and \$52,000.
- The retail price of each handbag is \$65. Based on a small amount of sales data available, the demand has fluctuated between 2,500 and 5,000 handbags per quarter, but in most quarters the demand has been about 3,500 handbags.
- Conduct a risk analysis to assess the quarterly profitability of the company.

Assessing Risk

- Example continued with Excel
- Profit = (Retail price per handbag – Variable cost per handbag) × Demand – Fixed cost
-

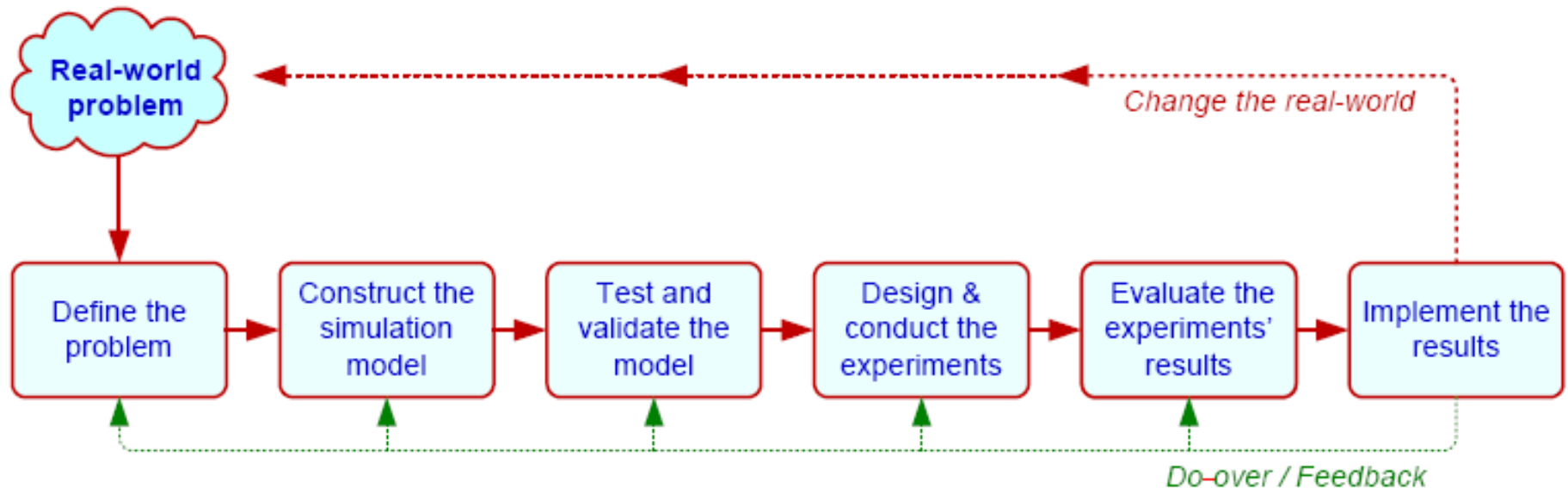
	A	B	C	D
1	Example 16.1: Risk analysis			
2				
3		Scenarios		
4	Input parameters:	Worst case	Most likely case	Best case
5	Retail price	\$65	\$65	\$65
6	Variable cost	\$46	\$40	\$35
7	Fixed cost	\$52,000	\$43,000	\$31,000
8	Demand	2,500	3,500	5,000
9				
10	Output:			
11	Profit calculation	(\$4,500)	\$44,500	\$119,000

Simulation

- To **imitate real-world business scenarios**.
 - for ***conducting experiments***
 - is normally used ***when a problem is too complex*** to be treated using numerical optimization techniques
- Simulations are often performed using computer software.
 - Study how changes in a certain variable might impact the process
 - Construct the simulation to adequately and accurately reflect the process
 - Quickly experiment with different values and parameters ***without disrupting the actual business processes***

Simulation Methodology

- Model real system and conduct repetitive experiments.
- Steps:
 1. Define problem
 2. Construct simulation model
 3. Test and validate model
 4. Design experiments
 5. Conduct experiments
 6. Evaluate results
 7. Implement solution



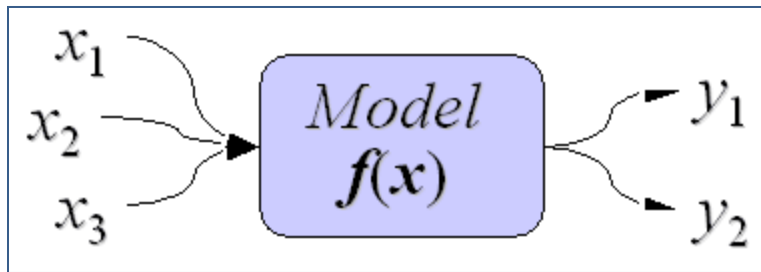
Simulation types

- Deterministic .vs. Stochastic
- Time-dependent .vs. Time-independent Simulation
- Discrete event .vs. Continuous simulation

Deterministic versus Stochastic

- **Deterministic model** is one whose behavior is entirely predictable. The system is determined using precise value of input and output
- **Stochastic model** is one whose behavior cannot be entirely predicted. The system is determined based on the predefined distributed pattern of input and the responses of output.

Simulation Type-- Deterministic versus Stochastic

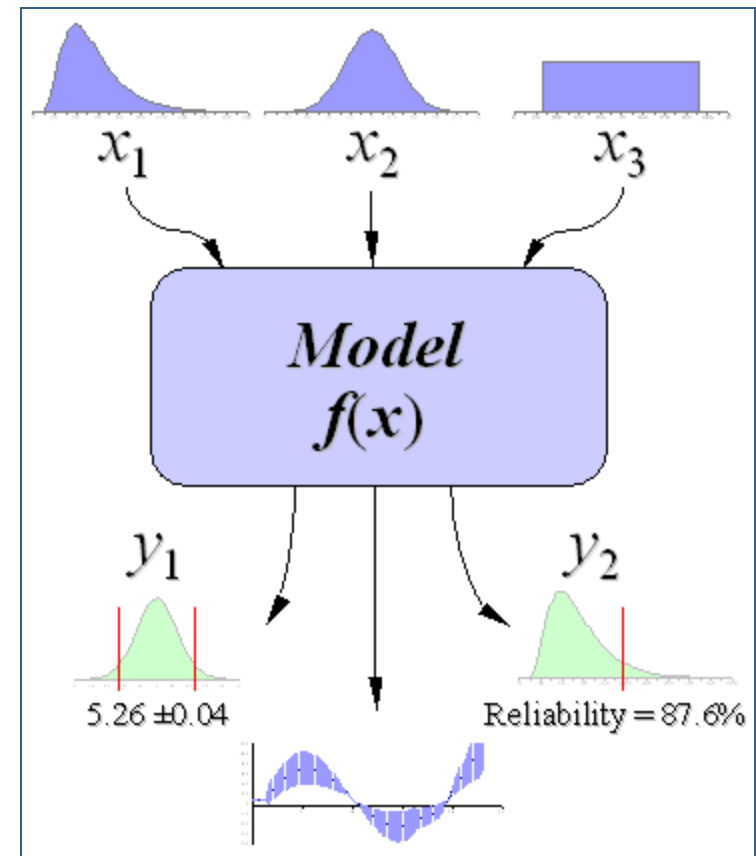


Deterministic model

Stochastic can be of discrete or continuous probabilistic distribution function:

e.g.

randbetween(a,b),
round(rand()),



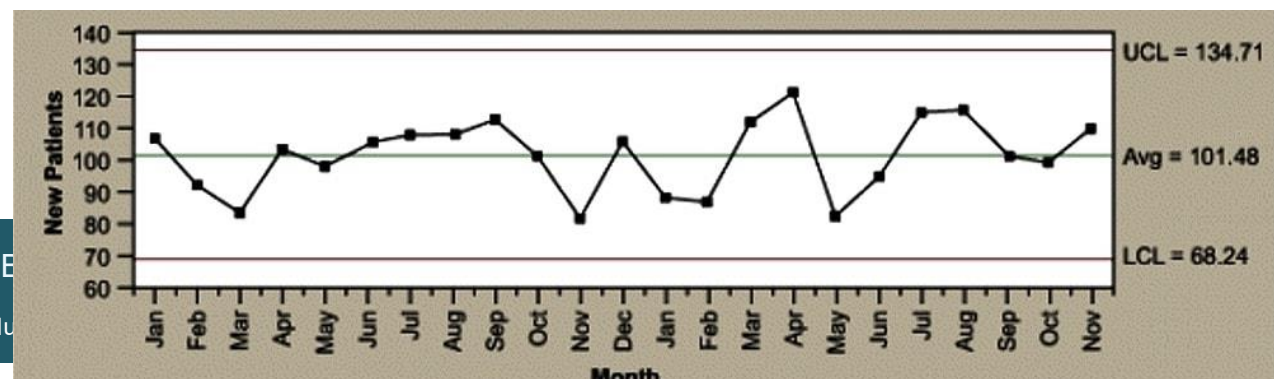
Stochastic model

Time-dependent .vs. Time-independent Simulation

- This refers to how important is the time to the problem when the event occurred.
- For example, if we do not care what time of the day the pattern of demand is, then this problem is time-independent.
- However, in some application like waiting-line in e-commerce system, it is crucial to know quite precise time of arrival (time-dependent situation.)

Discrete vs Continuous events

- **Discrete event** means the study of interaction between entities in the system whose state is discrete and changes at particular time point and then remains in that state for some time.
- The example is a server and customers in a shop.
 - The distributions of customer arrival and server ability can be modeled to observe the average performance of the system, e.g. no. of waiting customers.
- **Continuous** simulation is appropriate for systems with a continuous state that changes continuously over time.
- An example is the amount of liquid in a tank and or its temperature.
 - Such a system can be described by differential equations or visualization (e.g. control chart):

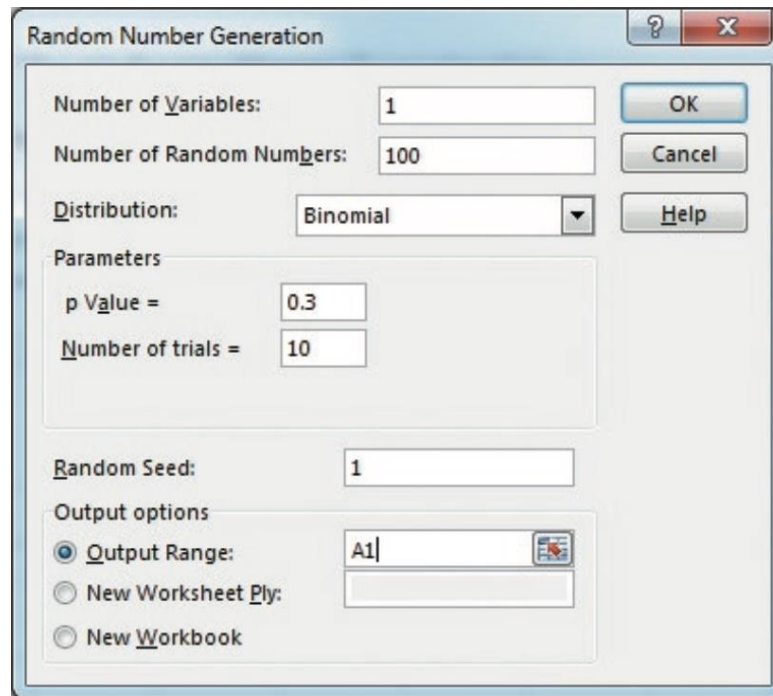


Random variables for assessing stochastic event

- Example: A company estimates that 30% of its customers would react positively to its new web features.
- Here, we demonstrate how to generate random observations that represent the number of customers who react positively to the company's new web features in a randomly selected group of 10 customers.
- Simulate random observations based on 100 groups of 10 customers.
 - This is a binomial distribution experiment

Assessing Risk

- Example continued with Excel
- Use BINOM.INV and RAND
- Data > Data Analysis > Random Number Generation



The image shows the 'Random Number Generation' dialog box in Microsoft Excel. The dialog box has a title bar with a question mark and a close button. It contains several input fields and buttons. The 'Number of Variables' is set to 1, and the 'Number of Random Numbers' is set to 100. The 'Distribution' is set to 'Binomial'. The 'Parameters' section shows 'p Value =' set to 0.3 and 'Number of trials =' set to 10. The 'Random Seed' is set to 1. The 'Output options' section has three radio buttons: 'Output Range:' (selected), 'New Worksheet Ply:', and 'New Workbook'. The 'Output Range:' is set to 'A1'.

Random Number Generation

Number of Variables: 1

Number of Random Numbers: 100

Distribution: Binomial

Parameters

p Value = 0.3

Number of trials = 10

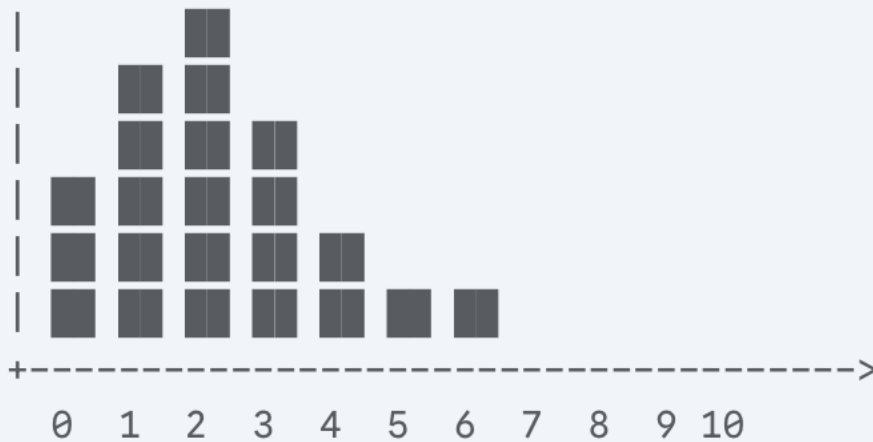
Random Seed: 1

Output options

☒ Output Range: A1

☐ New Worksheet Ply:

☐ New Workbook



Binomial distribution

- **Definition:** Prob. of getting a specific number of "successes" in a fixed number of independent trials.
- **Trial Types:** Each trial must have only two possible outcomes (e.g., success/failure, yes/no, heads/tails).
- **Fixed Trials:** The number of trials (n) is predetermined and constant.
- **Independence:** Each trial must be independent; the outcome of one trial does not affect the others.
- **Constant Probability:** The probability of success (p) must be the same for every single trial. The prob. of failure is $q=1-p$.
- **Discrete:** It's a *discrete* probability distribution, meaning the variable (number of successes) can only take specific integer values (0, 1, 2, ..., n).

Appropriate probability distribution of a random variable

- In many real-world settings, the appropriate probability distribution of a random variable is not known in advance.
- **Approach 1:** Plot the data and inspect the histogram for distinctive shapes corresponding to theoretical distributions.
- **Approach 2:** Construct and use an empirical probability distribution by using historical or empirical data.
 - An empirical probability distribution can be represented by a relative frequency table.
 - Cumulative relative frequencies are then used to generate random observations that reflect the pattern observed in the historical data.

Assessing Risk

- Example: A medical supply provider examines the weekly demand by hospitals for respiratory ventilators.

Ventilator demand	Frequency
100	20
200	25
300	35
400	10

- Construct 100 weekly demands for ventilators based on the empirical probability distribution.

Assessing Risk

- Example continued with Excel

	A	B	C	D	E
1	Ventilator demand	Frequency	Relative frequency	Probability range (lower)	Probability range (upper)
2	100	20	0.2222	0.0000	0.2222
3	200	25	0.2778	0.2222	0.5000
4	300	35	0.3889	0.5000	0.8889
5	400	10	0.1111	0.8889	1.0000

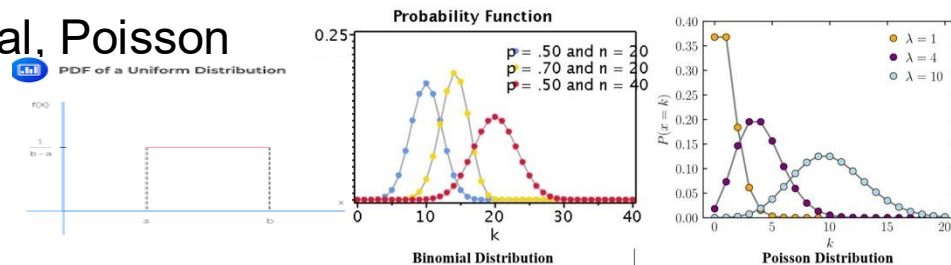
- Data > Data Analysis > Random Number Generation
 - Setting Distribution=uniform, values between 0 and 1.
 - Using Xlookup to find the simulated demand.

	G	H	I	J	K	L
1	Week	Random number	Simulated demand			
2	1	0.0013	100		Mean	237.00
3	2	0.5636	300		Standard deviation	93.9106
4	3	0.1933	100			

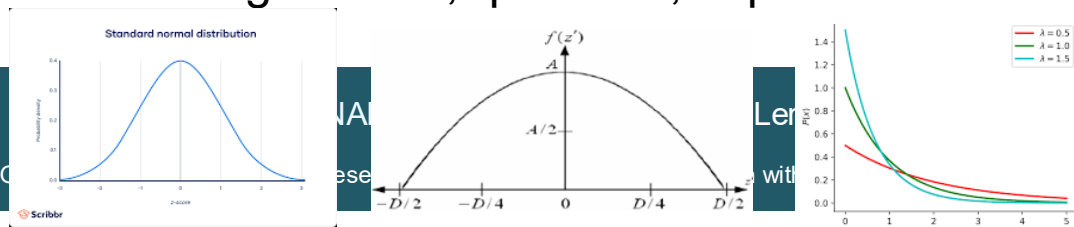
Monte Carlo Simulation

- A family of computer algorithms **used to model risk and/or uncertainty.**
 - Developed by Stanislaw Ulam and John von Neumann
 - Studied nuclear fission on the Manhattan Project at Los Alamos National Laboratory during WWII
 - Named after the Monte Carlo Casino in Monaco, in reference to games of chance: a popular attraction.
- Used to *model probabilistic (stochastic) processes* to recreate real-world processes.
 - Random sampling to mimic the odds of all possible outcomes
 - Needs to capture the uncertainty of the relevant variables

- **Discrete:** e.g. uniform, binomial, Poisson



- **Continuous:** e.g. normal, quadratic, exponential



Monte Carlo simulation method

- is a method to solve the stochastic problem by **iteratively evaluating a deterministic model** using random inputs drawn from specified probability distributions.
 - Often used when the model is complex, nonlinear, or involves more than just a couple uncertain parameters
 - Deterministic = static inputs (means you get the same results no matter how many times you re-calculate)
 - Stochastic = dynamic inputs (e.g. represented by probabilistic distribution model)

Modeling by Monte Carlo

Just a simple 5 steps

- **Step 1:** Create a parametric model, $y = f(x_1, x_2, \dots, x_q)$.
- **Step 2:** Generate a set of random inputs, $x_{i1}, x_{i2}, \dots, x_{iq}$.
- **Step 3:** Evaluate the model and store the results as y_i .
- **Step 4:** Repeat steps 2 and 3 for $i = 1$ to n . n = number of simulated events
- **Step 5:** Analyze the results using histograms, summary statistics, confidence intervals, etc.

Exercise of Monte Carlo Simulation

An investor wants to invest \$300,000 in a portfolio of 3 mutual funds. The annual fund returns are normally distributed with a mean of 2.0% and Stdev of 0.3% for the short-term investment fund, a mean of 5.0% and Stdev of 2.5% for the intermediate-term fund, and a mean of 6.25% and stdev of 5.5% for the long-term fund. An initial plan for the investment allocation is 45% in the short-term, 35% in the intermediate-term, and 20% in the long-term fund.

- a) Use a seed of 1, to simulate 100 trials to estimate the mean ending balance after the first year and assess the risk of this investment.
- b) What if the allocation is change to 30:55:15?

Advantages of Simulation

- The theory is fairly straight forward
- Great deal of time compression
- Can include the real complexities of problems
- Can handle wide variety of problem types
- The model reflects manager's perspective
- Produces important performance measures
- Often it is a quick DSS modeling tool for non-structured and stochastic problems

Limitations of Simulation

- Cannot guarantee an optimal solution
- Slow and costly construction process
- Cannot transfer solutions and inferences to solve other problems (problem specific)
- So easy to explain/sell to managers, may lead overlooking analytical solutions

End

- Concept of Simulation as a method for Prescriptive analysis
 - We have learnt the concept, tools and application for Simulation

Further readings

- Some practical guideline for conducting the simulation process is available at <http://analytics-magazine.org/spring-2009/205-software-solutions-the-abcs-of-simulation-practice.html>
- Monte Carlo application
<http://www.vertex42.com/ExcelArticles/mc/SalesForecast.html> or
<https://www.riskamp.com/how-to>.